



Hochschule München

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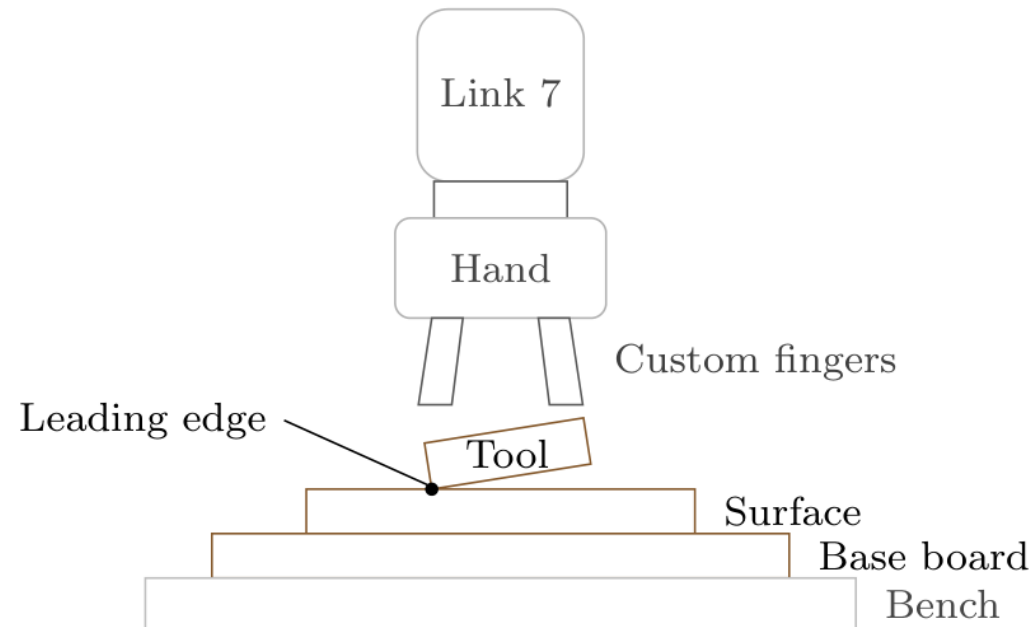
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

Main thesis question

- **How do stiffness and the compliance centre affect passive alignment?**

Idea

- Apply the press; let contact rotate the tool

1. Cartesian impedance and centre of compliance
2. Contact experiments and results
3. Null-space study
4. Conclusion and future work

Cartesian wrench = force and moment together

Position and force

Desired / measured p_d, p_{EE}

$$e_p = p_d - p_{EE}$$

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

K_p, K_R translational / rotational stiffness

$\dot{p}, \omega$ linear / angular velocity

Orientation and moment

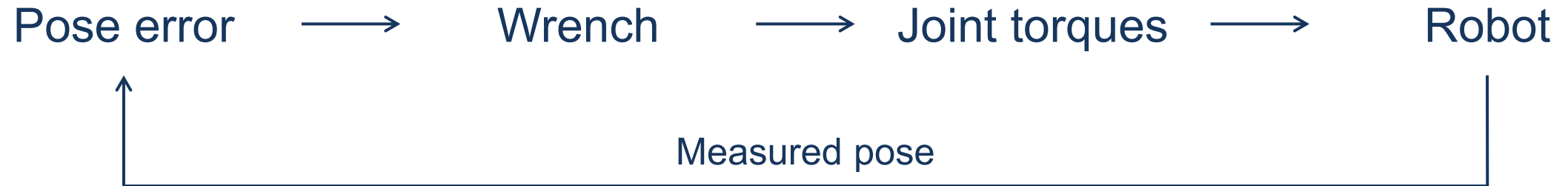
Desired / measured R_d, R_{EE}

e_R rotational correction

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

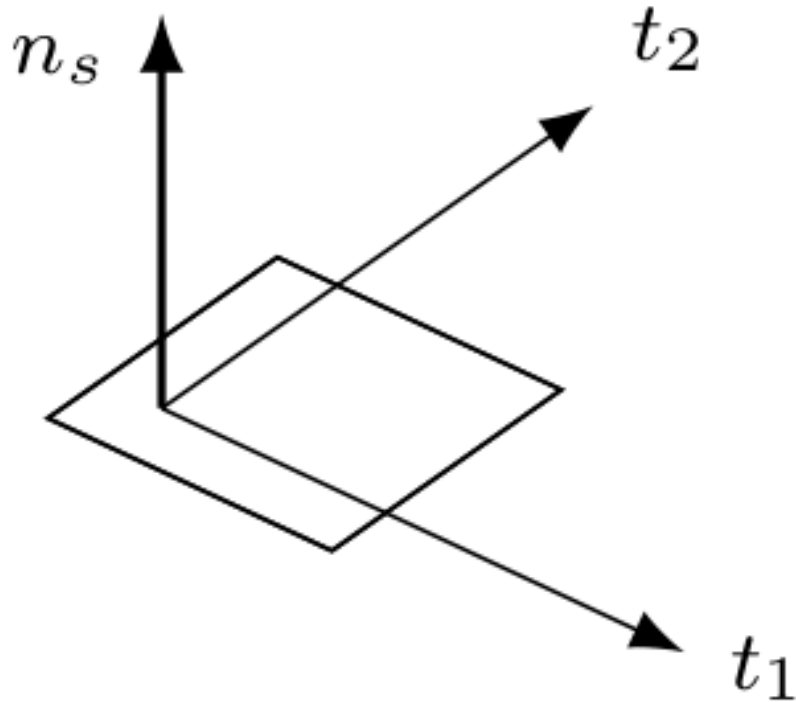
D_p, D_R translational / rotational damping

d, EE desired / measured end effector



$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

1 kHz control loop · C++ / libfranka



- n_s outward surface normal
- t_1, t_2 two perpendicular surface tangents

Translation

- Softer normal pressing
- Stiffer motion along the surface

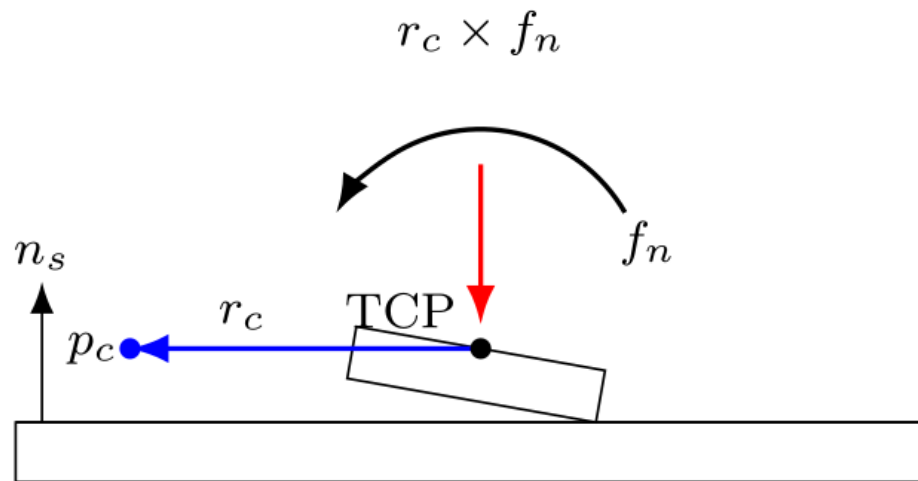
Rotation

- Softer rotations about the tangents
- Stiffer rotation about the normal

Virtual reference point

p_c CoC position

r_c Displacement from TCP to CoC

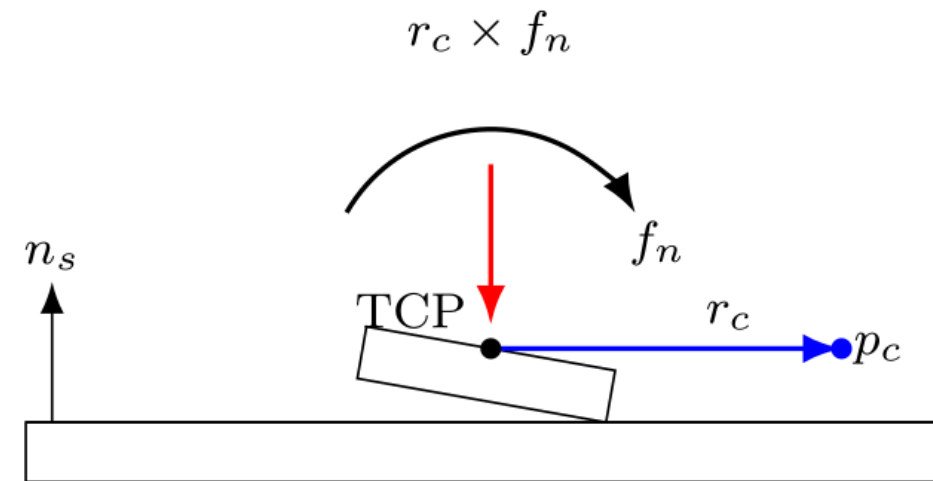


(a) Supporting displacement

Commanded TCP moment

$$m = m_R + r_c \times f$$

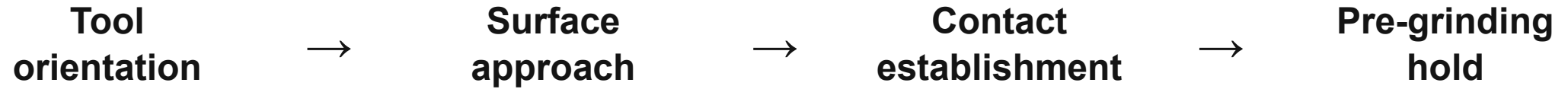
m_R rotational spring-damper moment



(b) Reversed displacement

- Opposite displacement reverses the added moment

Experimental procedure

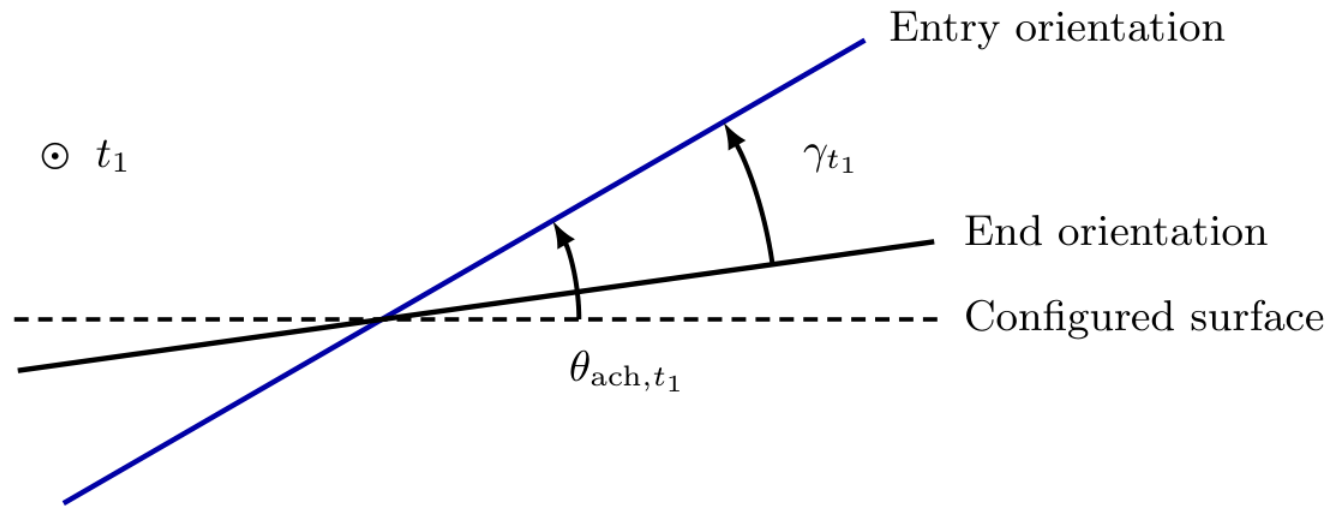


Contact study

- Rectangular tool: 40 × 120 mm
- Five-second contact phase
- Fixed entry orientation reference

Controlled comparison

- A: centre at the TCP
- B / C: rotational / translational stiffness
- D: tangential compliance-centre position
- 57 trials; three repetitions per setting

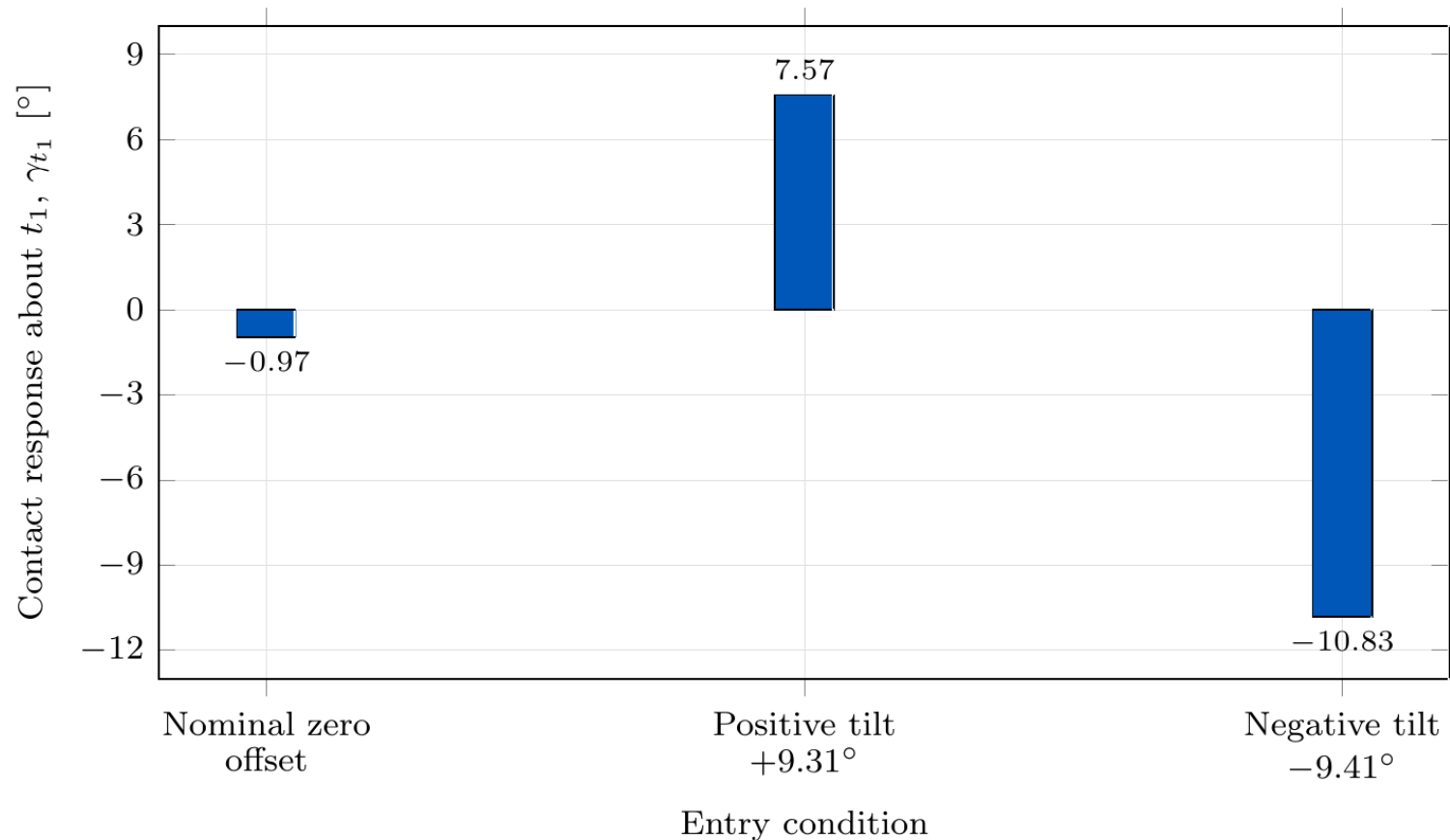


θ_{ach,t_1} Achieved tilt
Orientation at contact entry

γ_{t_1} Contact response
Rotation relative to the held
entry orientation

- Aligning response: same sign as the entry tilt
- Calculated from measured end-effector orientation

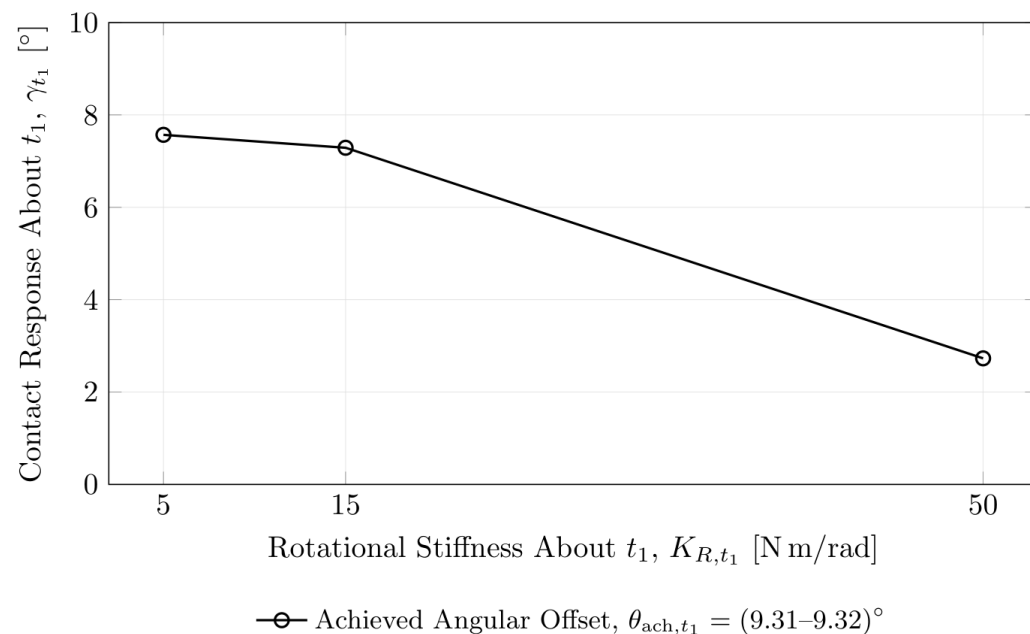
Baseline contact response



- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Rotational stiffness

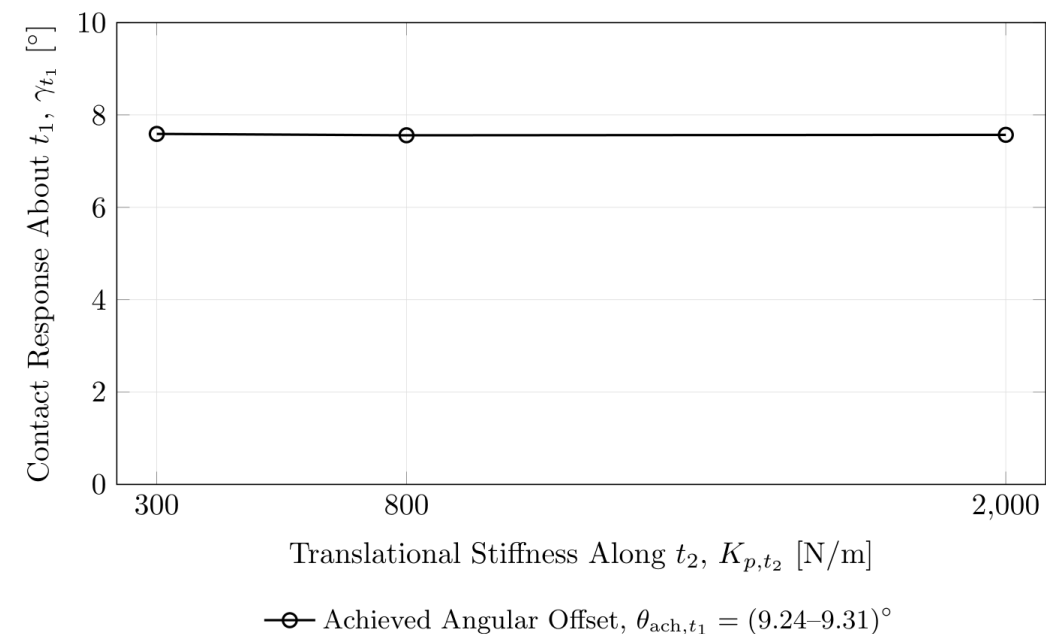
Resistance to rotation about t_1



- Rotational stiffness: 63.9% lower response

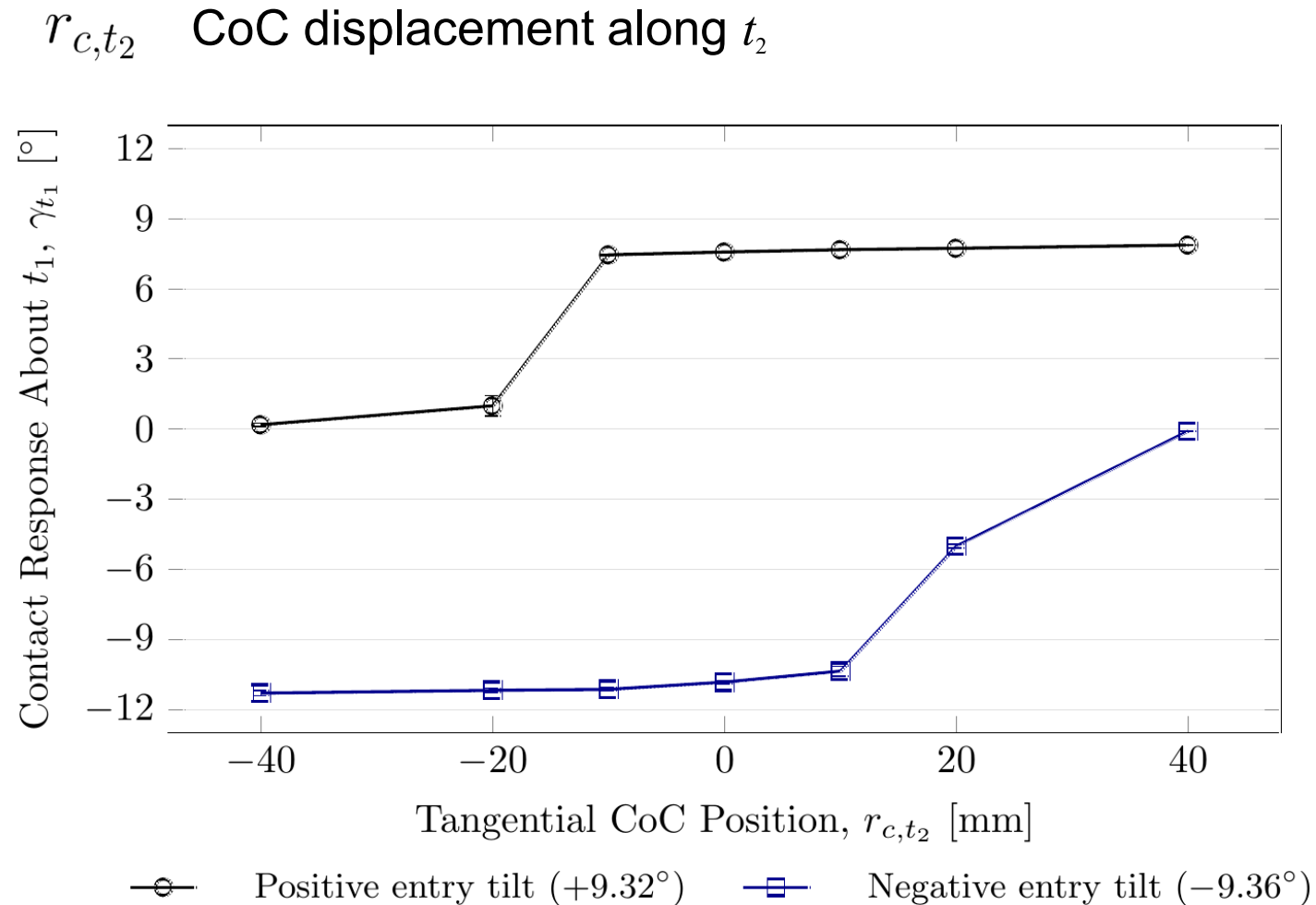
Translational stiffness

Resistance to sliding along t_2



- Tangential stiffness: 0.4% response span

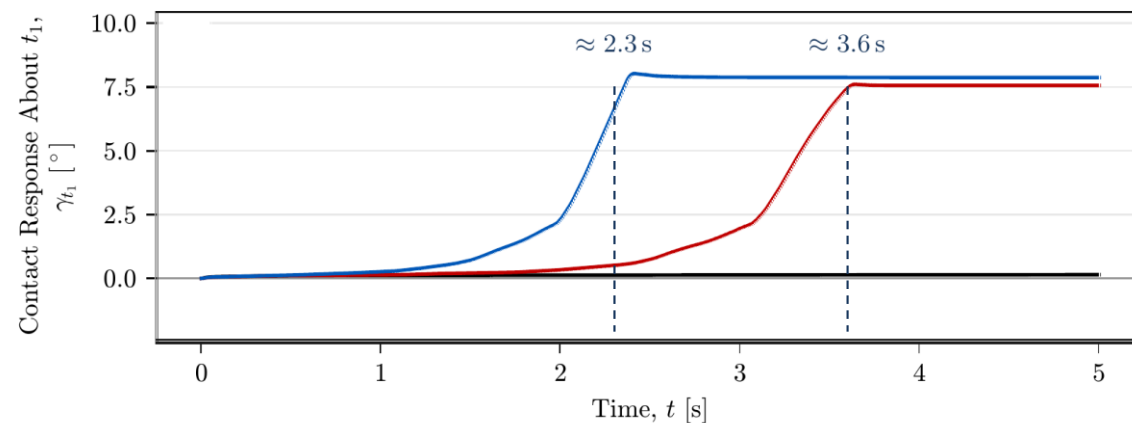
Effect of CoC position



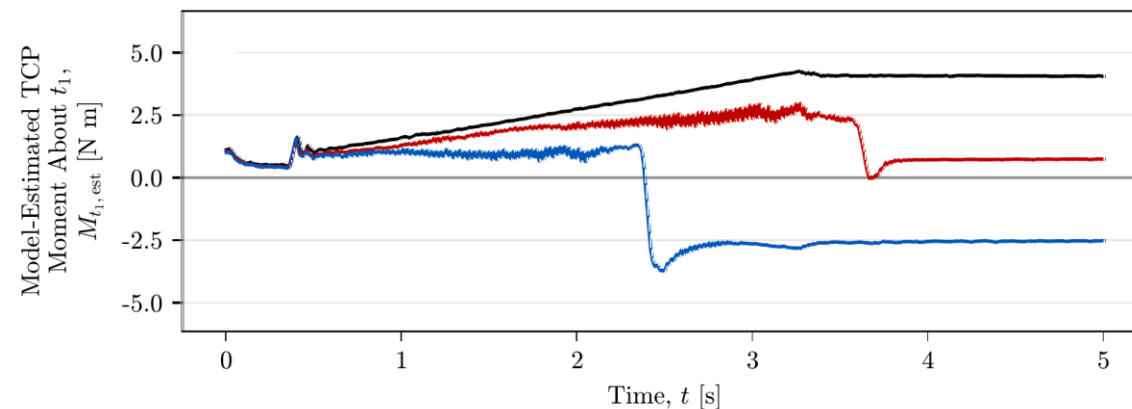
Relative to the TCP-centred baseline

- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

Contact response



Interaction moment at the TCP



— CoC Position, $r_{c,t_2} = -40$ mm — CoC at TCP, $r_{c,t_2} = 0$ — CoC Position, $r_{c,t_2} = 40$ mm

- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces
- Final normal force (approx.): -83 N (-40 mm), -79 N (TCP), -78 N ($+40$ mm)

Contact demonstration



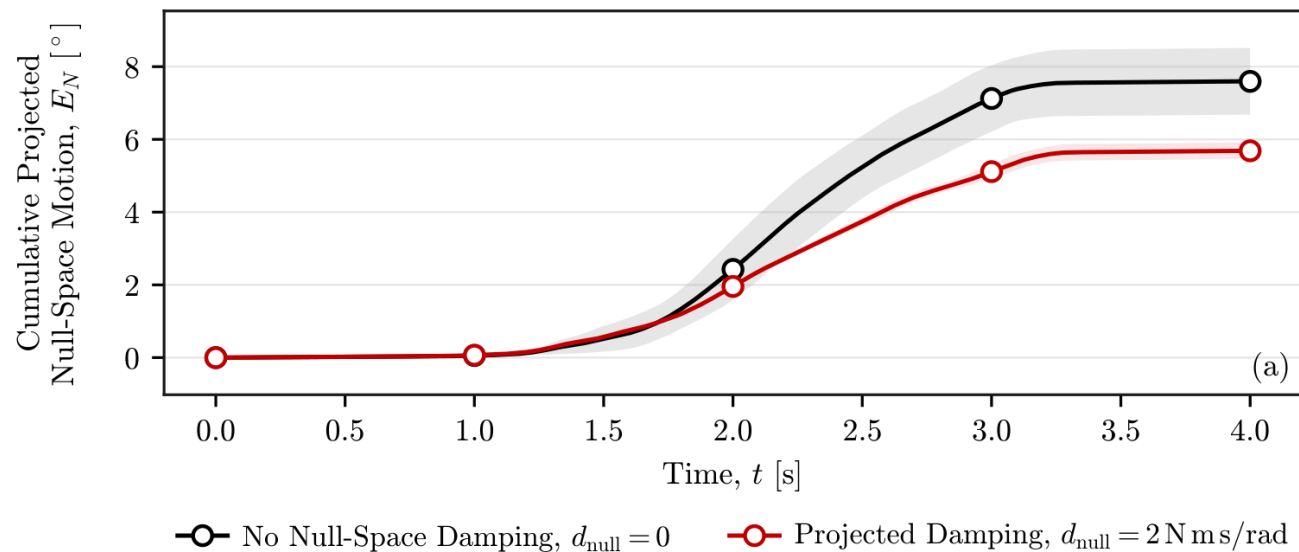
Pre-grinding hold · fixed desired orientation · CoC at TCP

Secondary study: null-space motion

7 joints, 6-DOF tool pose: coordinated motion with the pose held

Damping

Accumulated projected motion



- 25% less accumulated motion
- Maximum TCP position error: 1.304 mm (criterion: 2 mm)

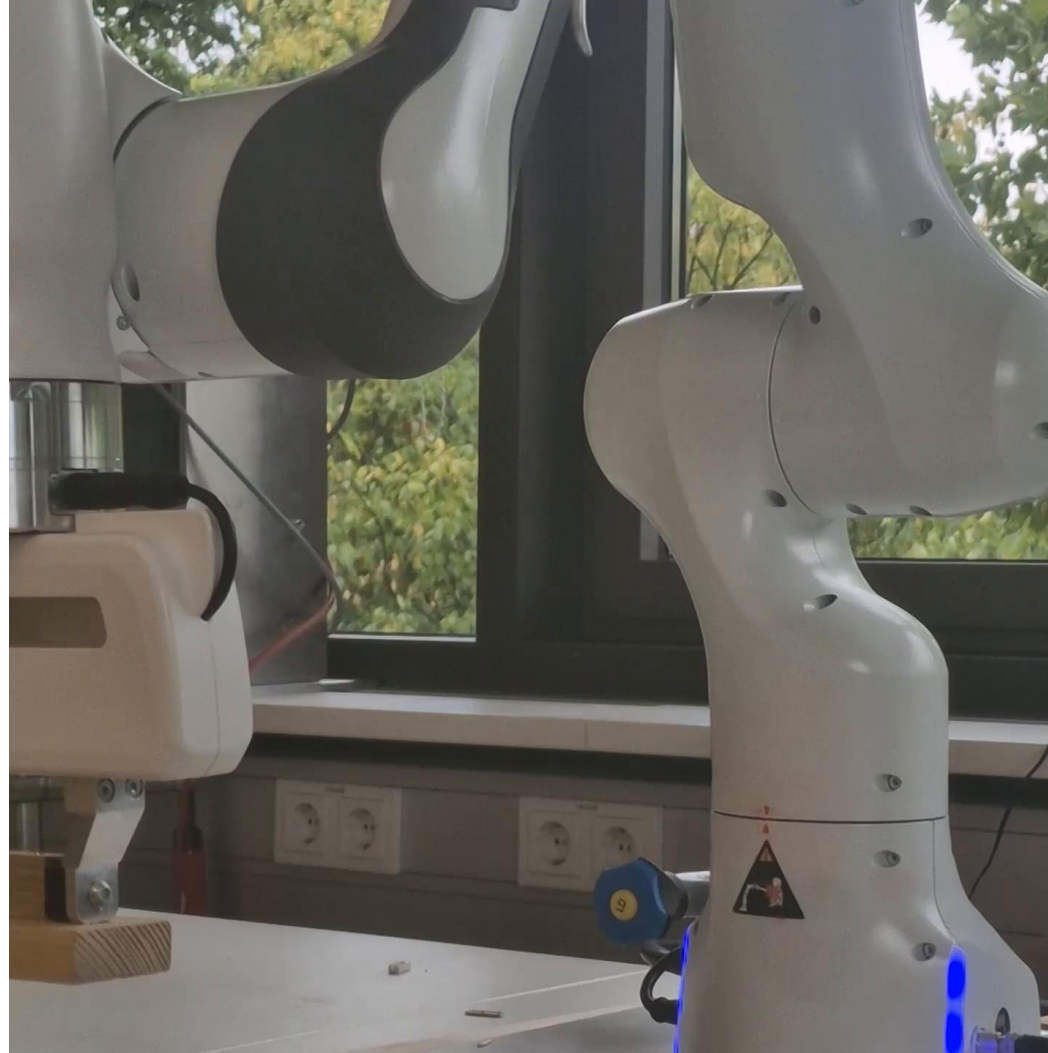
Conditioning

Mean net projected displacement

Setting	Mean (°)
No null-space torque	7.517
Damping	5.609
Conditioning, gain 1.5	0.015
Conditioning, gain 2	-0.006

- Near-zero net displacement
- Stronger gain: more back-and-forth motion

Disturbance demonstration



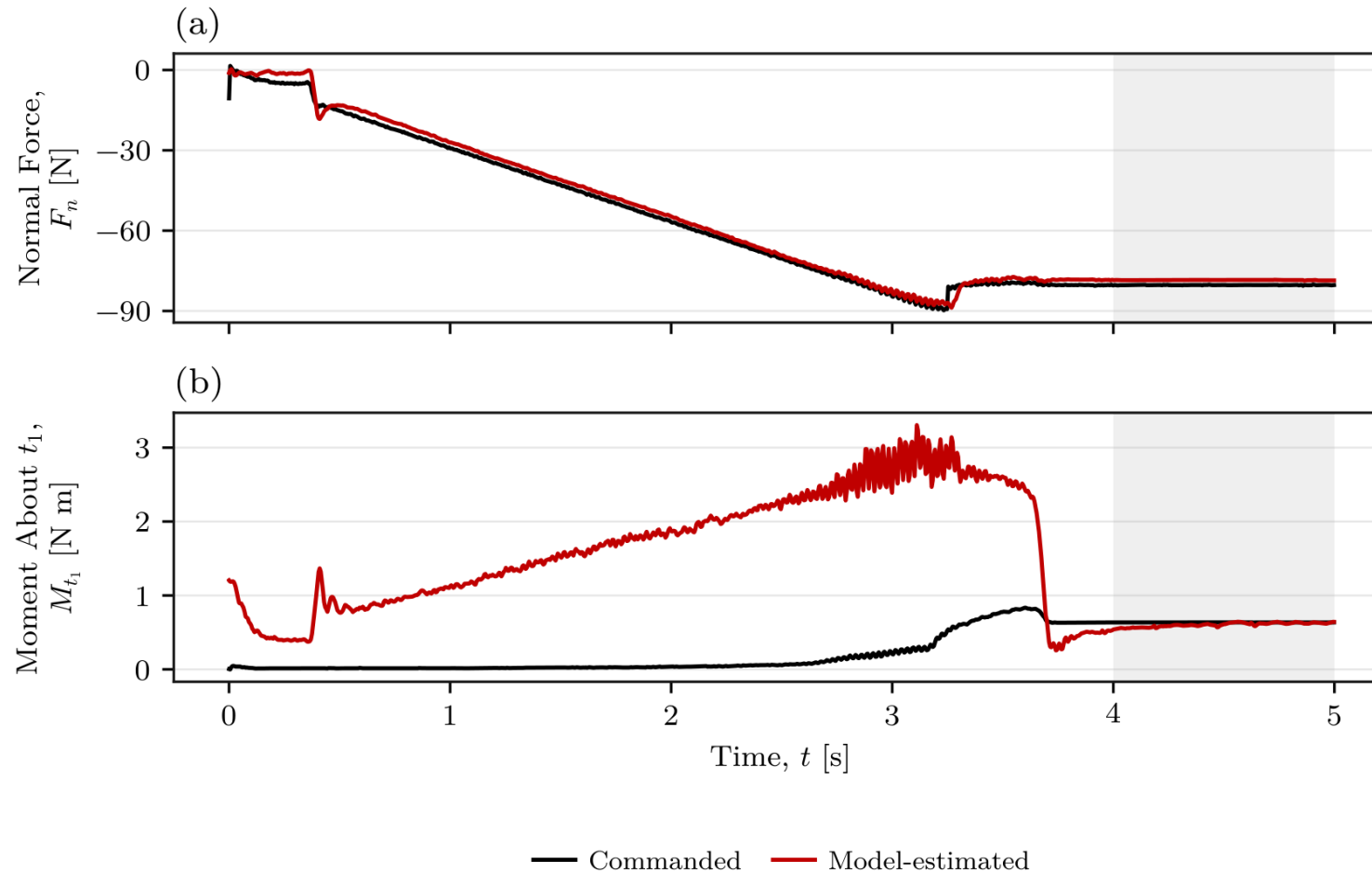
No null-space torque → Damping → Conditioning

- Passive alignment through Cartesian impedance
- Rotational stiffness controls response magnitude
- CoC displacement selects a preferred rotation direction

Future work

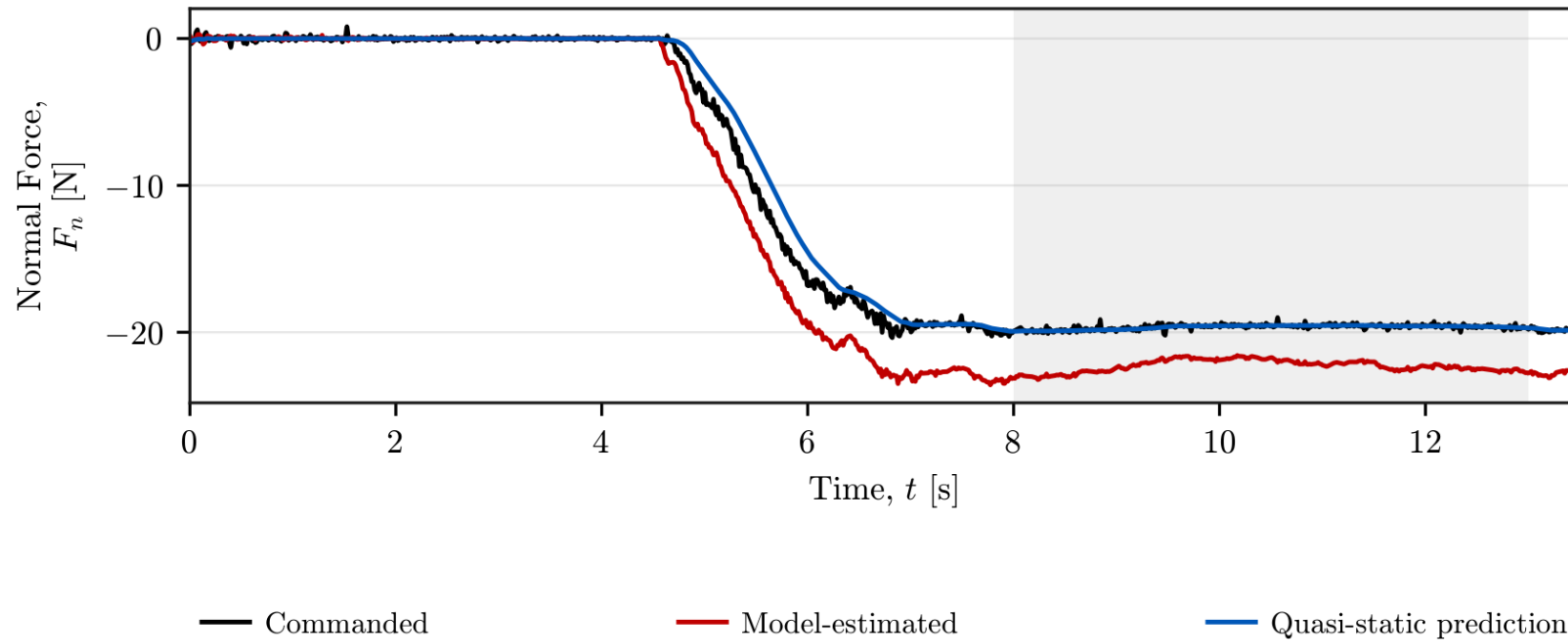
- Tilt-dependent CoC selection
- Return to the TCP after alignment
- Direct angle measurement and sustained grinding

Commanded and estimated wrench



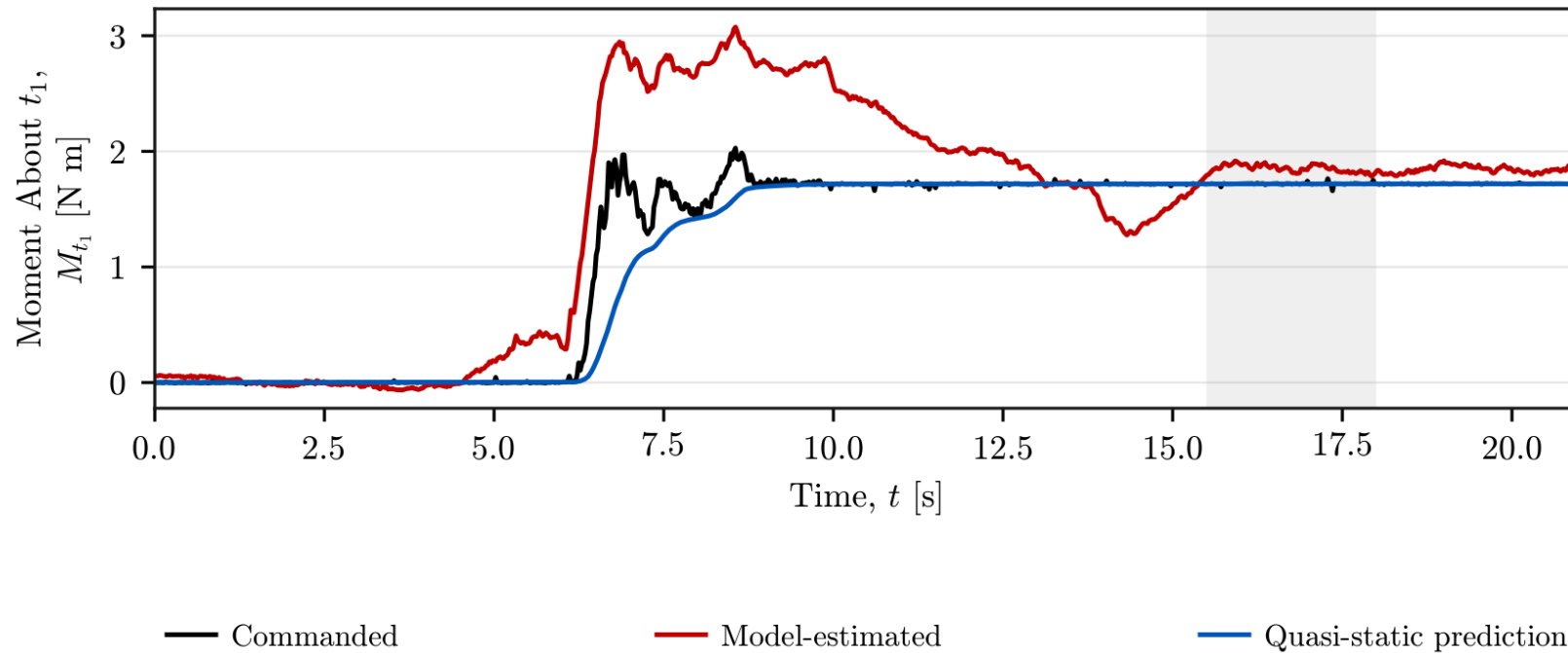
- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$



- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{EE} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{EE} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

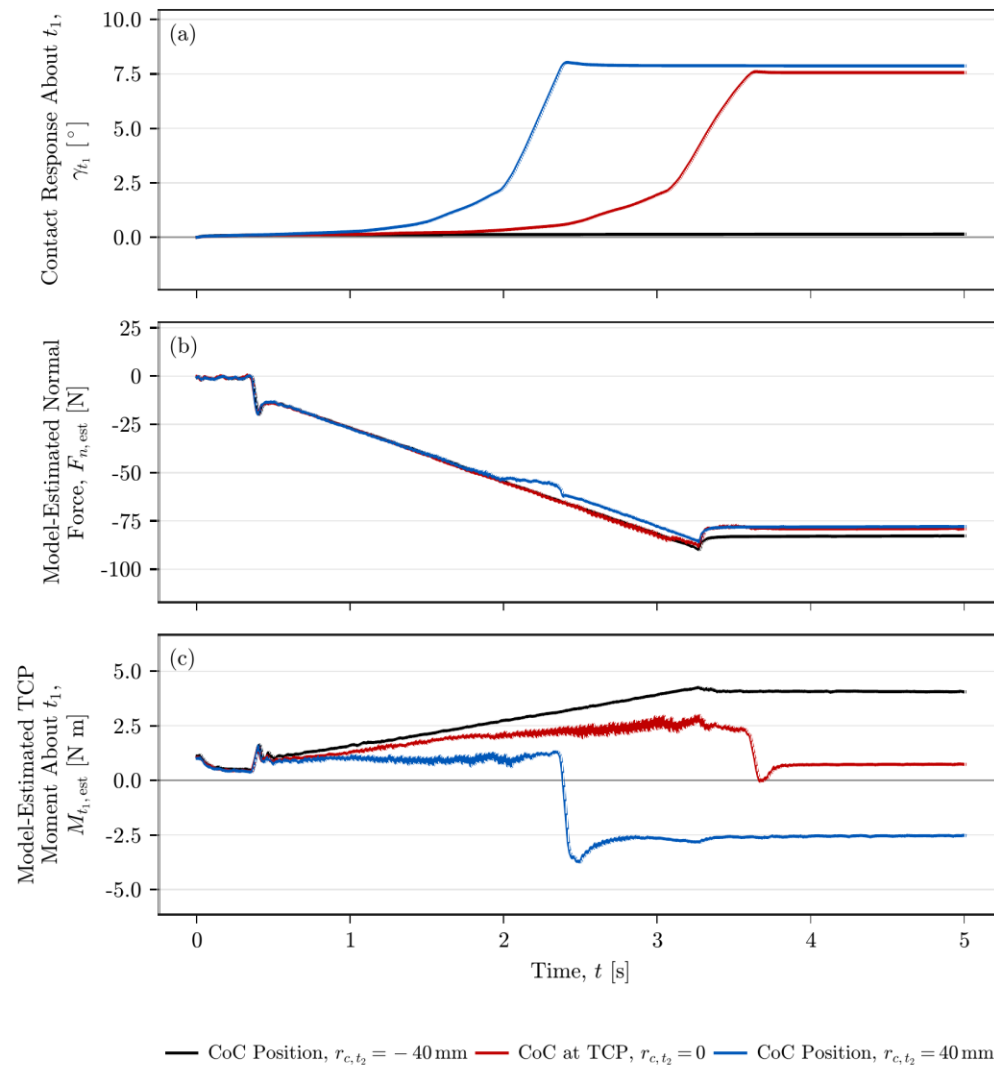
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



Representative traces
Positive entry tilt

- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates